Commercial Banking Customer Transaction & Risk Analytics

(Ethiopian Banking Sector Simulation)

1. Project Overview

This project analyzes customer transaction behavior, financing portfolio performance, digital banking adoption, and credit risk patterns using a dataset of **15,000 records** modeled after real-world Ethiopian commercial banking operations.

The project bridges data engineering and financial intelligence to simulate how modern financial institutions leverage data analytics to optimize lending portfolios, mitigate defaults, detect potential fraud, and drive digital transformation.

Tech Stack & Tools

- **Data Prep & Feature Engineering:** Python (`Pandas`, `NumPy`, `SQLAlchemy`)
- **Data Warehousing & Business Intelligence:** PostgreSQL, SQL
- **Data Visualization & Executive Reporting:** Power BI
- **Version Control & Documentation:** GitHub

2. Dataset Summary

- **Rows:** 15,000
- **Data Grain:** One row per customer record with aggregated transaction and financing status.
- **Core Dimensions & Features:**
  - **Demographics:** `customer_id`, `age`, `gender`, `city`, `occupation`, `monthly_income`, `relationship_years`, `customer_segment` (Retail, SME, Corporate, Agricultural, Diaspora, Women, Youth).
  - **Transactional Activity:** `transaction_type`, `transaction_amount`, `channel`, `mobile_app_usage`, `monthly_login_frequency`, `branch_name`.
  - **Financing Portfolio:** `loan_amount`, `loan_type`, `loan_status` (Paid, Active, Late Payment, Defaulted), `days_past_due`, `approval_status`.
  - **Risk & Compliance:** `collateral_type`, `sector_financed`, `kyc_status`, `internal_risk_rating`.

3. Exploratory Data Analysis using Python

Data cleaning, validation, and schema preparation were conducted programmatically in Jupyter Notebooks:

- **Initial Profile:** Inspected structural integrity via `df.info()` and statistical distributions using `df.describe()`.

	customer_id	age	monthly_income	transaction_amount
count	15000.000000	15000.000000	15000.000000	15000.000000
mean	17500.500000	40.979267	29030.398733	62697.961800
std	4330.271354	13.478937	20236.508849	55356.403878
min	10001.000000	18.000000	4001.000000	1074.000000
25%	13750.750000	30.000000	14079.750000	1074.000000
50%	17500.500000	41.000000	22087.500000	51063.000000
75%	21250.250000	53.000000	39072.750000	135055.000000
max	25000.000000	64.000000	99919.000000	135055.000000

	loan_amount	repayment_delay_days	credit_score
count	15000.000000	15000.000000	15000.000000
mean	250569.593333	59.407267	2.638800
std	143191.315621	34.769964	1.235436
min	65.000000	0.000000	1.000000
25%	127567.000000	29.000000	2.000000
50%	250509.500000	59.000000	2.000000
75%	372403.750000	89.000000	3.000000
max	499966.000000	119.000000	5.000000

	monthly_login_frequency	days_past_due	relationship_years
count	15000.00000	15000.000000	15000.000000
mean	17.08700	89.567333	7.992333
std	10.12519	52.233530	4.293037
min	0.00000	0.000000	1.000000
25%	8.00000	44.000000	4.000000
50%	17.00000	89.000000	8.000000
75%	26.00000	135.000000	12.000000
max	34.00000	180.000000	15.000000

- **Imputation & Deduplication:** Managed missing values using forward-filling methods and dropped redundant rows via `df.drop_duplicates()`.
- **Schema Standardization:** Transformed all column names to standard database `snake_case`.
- **Feature Engineering:** * Binned customer ages into structural economic cohorts (`Young Adult`, `Adult`, `Mid-Career`, `Senior`).
    - Aggregated transaction frequency and risk indicators into analytical base tables.
- **Database Pipeline:** Created a connection engine using `SQLAlchemy` to pipeline the data from a local Python environment straight into an enterprise PostgreSQL database layer.

4. Data Analysis using SQL (Business Queries)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Branch Transaction Performance

- Compares transaction performance across branches to identify high-performing regional hubs and support the expansion of effective localized operational strategies.

	branch_name text	total_transaction_volume numeric
1	Adama Branch	123564730
2	Bahir Dar Branch	119488502
3	Megenagna Bra...	119366920
4	Addis Main Bran...	118661236
5	Jimma Branch	117974987
6	Hawassa Branch	116242666
7	Bole Branch	115591590
8	Dire Dawa Branch	109578796

2. Digital Banking Engagement Analysis

- Segments customers based on digital banking usage to identify highly engaged users and optimize targeting strategies for digital financial products and app-exclusive services.

	customer_id bigint	monthly_login_frequency bigint
1	12859	34
2	12942	34
3	10296	34
4	10209	34
5	14899	34
6	14041	34
7	21528	34
8	23037	34
9	22561	34
10	12276	34

Total rows: 15000 Query complete 00:00:00

3. Credit Portfolio Exposure by Customer Segment

❖ Identifies customer segments with the highest average financing exposure, helping the bank monitor credit concentration and support balanced portfolio diversification.

	customer_segment text	average_loan_amount numeric
1	Youth Banking	256399.870041039672
2	Agricultural	254272.836257309942
3	Corporate	253209.510763209393
4	Women Banking	253024.913212435233
5	Retail	251480.704649390244
6	Diaspora	246077.366806136681
7	SME	245273.690942796610

4. Portfolio Default Vulnerability by Financing Type

- Identifies financing products with the highest default risk, enabling the bank to strengthen underwriting standards, improve portfolio risk management, and reduce potential credit losses.

	loan_type text	defaulted_loans bigint
1	Trade Financing	258
2	Agricultural Loan	251
3	Vehicle Financing	239
4	Murabaha Financi...	237
5	SME Financing	235

5. Concentration Risk by Economic Sector

- Evaluates financing portfolio exposure across economic sectors to identify areas of concentration risk and support portfolio diversification strategies in alignment with central bank regulatory requirements.

	sector_financed text	total_financing numeric
1	Transportation	397285206
2	Export Trade	383168074
3	Agriculture	380059209
4	Education	378552712
5	Import Trade	376889689
6	Health Services	376591334
7	Construction	372717682
8	Retail Trade	368989287
9	Hospitality	366926904
10	Manufacturing	357363803

6. Collateral Performance Assessment

- Evaluates the relationship between collateral types and loan default performance to measure liquidation reliability, strengthen asset-backed lending policies, and improve collateral risk management.

	collateral_type text	default_count bigint
1	No Collateral	185
2	Machinery	183
3	Personal Guarant...	178
4	Land	176
5	Building	173
6	Vehicle	167
7	Inventory	158

7. Suspicious Transaction & Anti-Money Laundering (AML) Detection

- Identifies unusually large withdrawal transactions compared with normal customer activity to generate early-stage compliance alerts and support timely Anti-Money Laundering (AML) monitoring and investigation.

	customer_id bigint	age bigint	gender text	city text	occupation text	monthly_income bigint	account_type text	transaction_type text	transaction_amount bigint	channel text	loan_status text
1	10013	53	Female	Addis Aba...	Government Employ...	16882	Mudarabah Savin...	Withdrawal	135055	POS	Active
2	10020	61	Female	Hawassa	Small Business Ow...	73987	Wadiah Deposit	Withdrawal	135055	Branch	Paid
3	10024	38	Male	Addis Aba...	University Lecturer	26173	Wadiah Deposit	Withdrawal	135055	Branch	Active
4	10055	25	Female	Dire Dawa	Teacher	18900	Wadiah Deposit	Withdrawal	135055	Mobile Banki...	Paid
5	10062	21	Male	Bahir Dar	Teacher	18331	Mudarabah Savin...	Withdrawal	135055	Mobile Banki...	Late Paymer
6	10064	23	Female	Jimma	Teacher	13402	Current Account	Withdrawal	135055	Mobile Banki...	Paid
7	10066	21	Female	Dessie	Private Employee	28678	Current Account	Withdrawal	135055	Mobile Banki...	Active
8	10088	42	Female	Addis Aba...	Teacher	8581	Savings Account	Withdrawal	135055	ATM	Paid
9	10097	26	Male	Bahir Dar	Teacher	15487	Salary Account	Withdrawal	135055	Mobile Banki...	Active
10	10146	30	Female	Addis Aba...	Private Employee	23770	Savings Account	Withdrawal	135055	POS	Paid
11	10154	56	Female	Bahir Dar	Merchant	45423	Savings Account	Withdrawal	135055	Mobile Banki...	Active
12	10155	62	Female	Adama	Teacher	19674	Mudarabah Savin...	Withdrawal	135055	Mobile Banki...	Paid
13	10161	49	Male	Hawassa	Small Business Ow...	20418	Salary Account	Withdrawal	135055	ATM	Active

8. Credit Delinquency Overdue Analysis

- Identifies financed sectors with the highest average repayment delays, enabling the bank to pinpoint slow-paying sectors and proactively prioritize risk monitoring, loan provisioning, and collection strategies.

	sector_financed text	average_days_past_due numeric
1	Manufacturing	92.1662068965517241
2	Transportation	91.5320754716981132
3	Education	90.9327505043712172
4	Agriculture	90.1024804177545692
5	Import Trade	89.9234863606121091
6	Construction	89.7544581618655693
7	Export Trade	89.4816053511705686
8	Health Services	87.6666666666666667
9	Retail Trade	87.5646349631614200
10	Hospitality	86.4966487935656836

9. High-Value Customer (HVC) Identification

- Identifies customers who generate the highest transaction value for the bank, enabling the prioritization of high-value clients for personalized relationship management, retention, and cross-selling opportunities.

	customer_id bigint	total_transaction_value numeric
1	22344	135055
2	23723	135055
3	23535	135055
4	18474	135055
5	10792	135055
6	20310	135055
7	23986	135055
8	12251	135055
9	23468	135055
10	11158	135055
11	14913	135055
Total rows: 5040	Query complete 00:00:0C	

10. Macro Socio-Economic Income Segmentation

- Classifies customers into income groups to support personalized banking services, targeted marketing campaigns, and tiered cross-selling strategies.

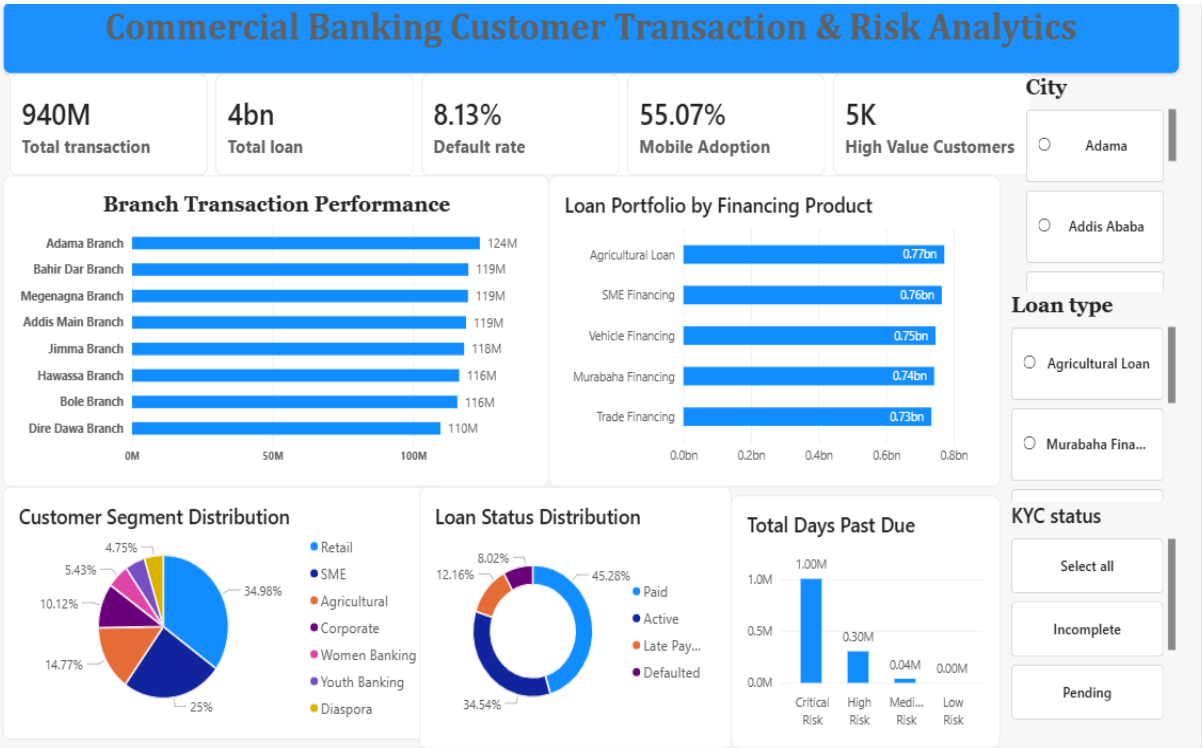
	customer_id bigint	income_segment text
1	10001	Middle Income
2	10002	Middle Income
3	10003	High Income
4	10004	Middle Income
5	10005	Middle Income
6	10006	Middle Income
7	10007	Middle Income
8	10008	Middle Income
9	10009	Middle Income
10	10010	High Income
11	10011	Middle Income
12	10012	Middle Income
13	10013	Middle Income
14	10014	High Income
15	10015	Low Income
16	10016	Middle Income

Total rows: 15000 Query complete 00:

5. Executive Dashboard in Power BI

Instead of sprawling multi-page views that cause dashboard fatigue, this project utilizes a hyper-focused **One-Page Executive Dashboard** mapping out the bank's core performance metrics at a glance.

- **Top-Level Key Performance Indicators (KPIs):** Surfaces Total Transactions (940M), Total Loan Portfolio (4B), Default Rate (8.13%), Mobile Adoption (55.07%), and High-Value Customer Count (5K).
- **Operational Performance Visuals:** Leverages clean horizontal bar charts to map out transaction performance by individual branches and credit exposures by financing products without truncating text.
- **Risk & Segment Distribution:** Integrates structured customer donut charts and clean column representations mapping out Days Past Due trends against specific internally rated risk metrics.



6. Business Recommendations

Portfolio Risk Management

- Strengthen underwriting for loan products with elevated default rates.
- Reduce concentration risk across sectors and financing products.

Digital Banking Strategy

- Increase digital banking adoption through targeted campaigns in low-engagement branches.
- Expand mobile banking services to reduce branch operating costs.

Customer Relationship Management

- Develop personalized programs for high-value customers.
- Increase cross-selling and customer retention initiatives.

Anti-Money Laundering (AML)

- Implement automated real-time alerts for unusually large withdrawal transactions.
- Enhance transaction monitoring to support regulatory compliance.

7. GitHub Repository Structure

Plaintext

```
Banking-Portfolio-Analytics/
│
├── data/
│   └── raw_bank_data.csv
│
├── python/
│   ├── data_cleaning.py
│   └── exploratory_analysis.py
│
├── sql/
│   └── banking_queries.sql
│
├── dashboard/
│   └── commercial_banking_dashboard.pbix
│
├── images/
│   └── executive_dashboard_screenshot.png
│
└── README.md
```